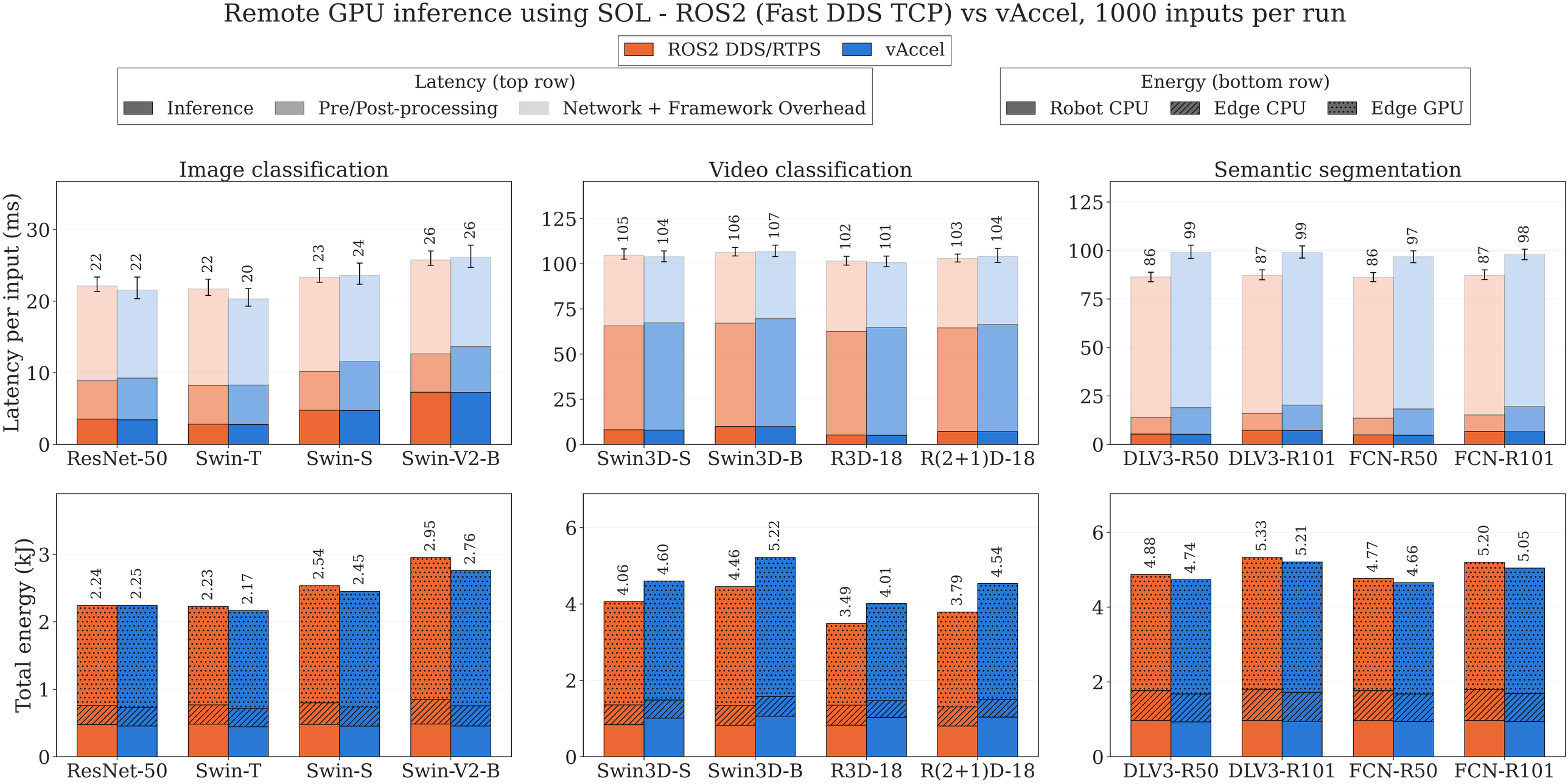

Remote GPU inference using SOL - ROS2 (Fast DDS TCP) vs vAccel, 1000 inputs per run
ROS2 DDS/RTPS
vAccel
Latency (top row)
Inference
Pre/Post-processing
Network + Framework Overhead
Energy (bottom row)
Robot CPU
Edge CPU
Edge GPU
Image classification
Video classification
Semantic segmentation
Latency per input (ms)
Total energy (kJ)
ResNet-50
Swin-T
Swin-S
Swin-V2-B
Swin3D-S
Swin3D-B
R3D-18
R(2+1)D-18
DLV3-R50
DLV3-R101
FCN-R50
FCN-R101
22 22 22 20 23 24 26 26
105 104 106 107 102 101 103 104
86 99 87 99 86 97 87 98
2.24 2.25 2.23 2.17 2.54 2.45 2.95 2.76
4.06 4.60 4.46 5.22 3.49 4.01 3.79 4.54
4.88 4.74 5.33 5.21 4.77 4.66 5.20 5.05